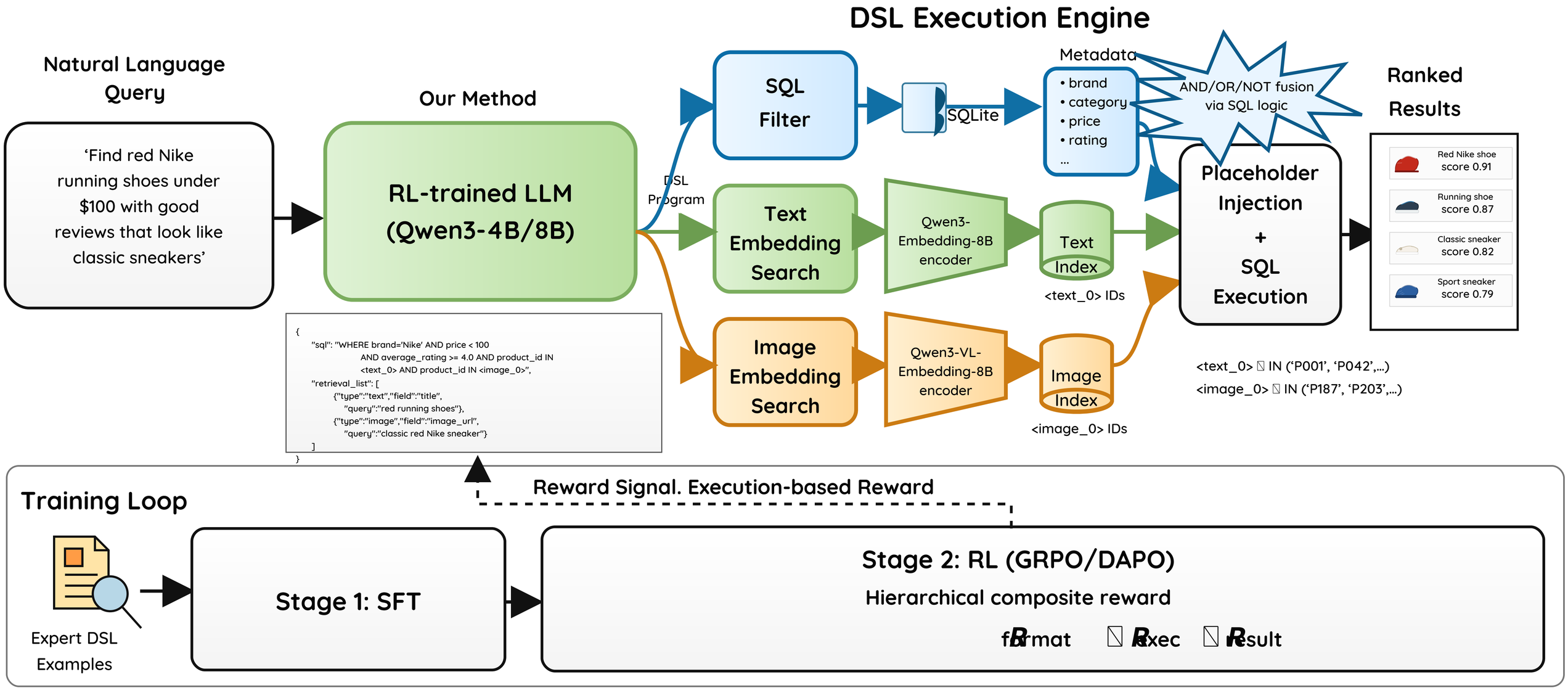
Natural Language Query
'Find red Nike running shoes under $100 with good reviews that look like classic sneakers'
Our Method
RL-trained LLM (Qwen3-4B/8B)
{
"sql": "WHERE brand='Nike' AND price < 100
AND average_rating >= 4.0 AND product_id IN
<text_0> AND product_id IN <image_0>",
"retrieval_list": [
{"type":"text","field":"title",
"query":"red running shoes"},
{"type":"image","field":"image_url",
"query":"classic red Nike sneaker"}
]
}
DSL Execution Engine
DSL Program
SQL Filter
SQLite
Metadata
• brand
• category
• price
• rating
...
Text Embedding Search
Qwen3-Embedding-8B encoder
Text Index
<text_0> IDs
Image Embedding Search
Qwen3-VL-Embedding-8B encoder
Image Index
<image_0> IDs
AND/OR/NOT fusion via SQL logic
Placeholder Injection + SQL Execution
<text_0> ∈ IN ('P001', 'P042',...)
<image_0> ∈ IN ('P187', 'P203',...)
Ranked Results
Red Nike shoe score 0.91
Running shoe score 0.87
Classic sneaker score 0.82
Sport sneaker score 0.79
Training Loop
Expert DSL Examples
Stage 1: SFT
Reward Signal. Execution-based Reward
Stage 2: RL (GRPO/DAPO)
Hierarchical composite reward
R_format ∧ R_exec ∧ R_result